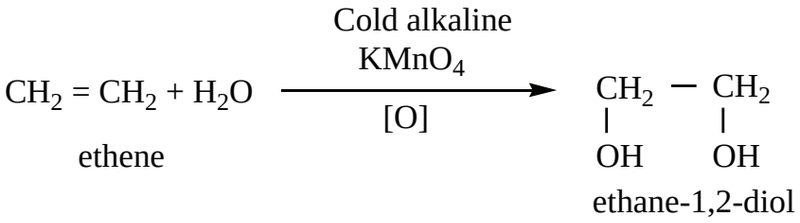
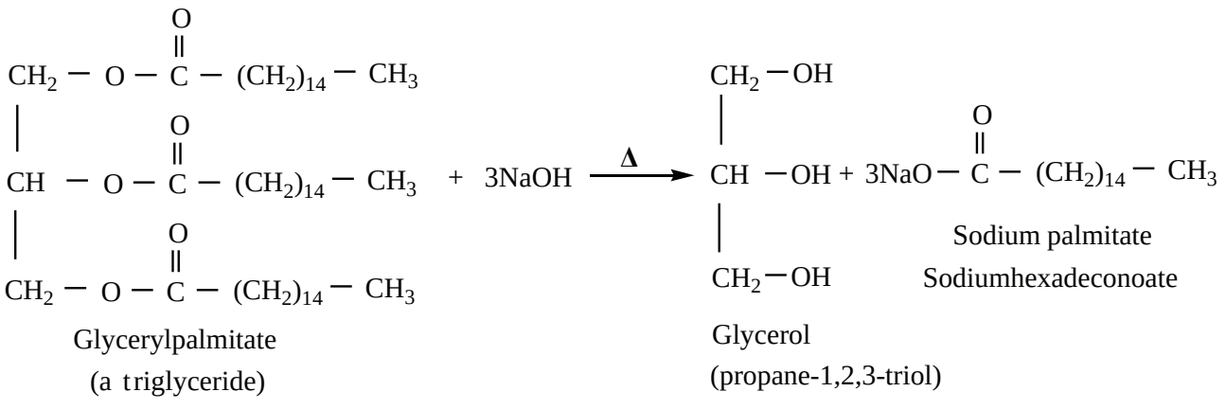
Preparation of glycol

We have already learnt that the hydroxylation of ethylene using cold alkaline solution of potassium permanganate (Baeyer's reagent) gives ethylene glycol.



Preparation of glycerol

Glycerol occurs in many natural fats and it is also found in long chain fatty acids in the form of glyceryl esters (Triglycerides). The alkaline hydrolysis of these fats gives glycerol and the reaction is known as saponification.



Evaluate Yourself?

1. Suggest a suitable carbonyl compound for the preparation of pent-2-en-1-ol using $LiAlH_4$.
2. 2-methylpropene $\xrightarrow{H_2SO_4/H_2O}$?
3. How will you prepare the following using Grignard reagent.
   i) t-butyl alcohol
   ii) allyl alcohol

Methods to differentiate primary, secondary and tertiary alcohols.

The following tests are used to distinguish between 1°, 2° and 3° alcohols.

a) Lucas test:

When alcohols are treated with Lucas agent (a mixture of concentrated HCl and anhydrous $ZnCl_2$) at room temperature, tertiary alcohols react immediately to form a turbidity due to the formation of alkyl chloride which is insoluble in the medium. Secondary alcohols react within 10 minutes to form a turbidity of alkyl chloride where primary alcohols do not react at room temperature.

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